INCH-POUND

MIL-T-26609C(USAF)
15 OCTOBER 1994
SUPERSEDING
MIL-T-26609B(USAF)
23 SEPTEMBER 1987

MILITARY SPECIFICATION

TRANSMITTER, POSITION

This specification is approved for use within the Departments of the Air Force and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 This specification covers the requirements for sealed position transmitters

1.2 Classification The transmitters shall be classified as follows

MS24550-1 - 2 1 Ratio
MS24550-2 - 1 1 Ratio

2 APPLICABLE DOCUMENTS

2 1 Government documents

2 1 1 Specifications, standards and handbooks The following specification, standards and handbooks form a part of this document to the extent specified herein Unless otherwise specified the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standard (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

Beneficial comments (recommendation, additions deletions) and any pertinent data which may be of use in improving this document should be addressed to Oklahoma City Air Logistics Center/ TICLA, 3001 Staff Drive, Tinker AFB OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method Of
MIL-S-7742	Screw Threads, Standard Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-T-31000	Technical Data Packages General Specification For
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810	Environmental Test Methods & Engineering Guidelines
MIL-STD-970	Standards And Specification Order Of Precedence For The Selection Of
MIL-STD-2073/1A	DOD Material Procedures For Development And Application Of Packaging Requirements
MS24550	Transmitter Position
MS33540	Safety Wiring And Cotter Pinning, General Practices For

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave Building #4, Section D, Philadelphia, PA 19111-5092)

2.2 Other publications The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation

American Society For Testing And Materials (ASTM)

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ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103)

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents These documents also may be available in or through libraries or other informational services)

2.3 Order of precedence In the event of a conflict between the text of this specification and the reference cited (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3 REQUIREMENTS

3 1 First Article When specified in the contract or purchase order a sample shall be subjected to first article inspection (see 4 5 and 6 4)

3 2 Selection of specifications and standards Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-970 except as provided in 3 2 1 and 3 2 2

3.2 1 Commercial parts Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts In any case, commercial utility parts, like screws bolts, nuts cotter pins, having suitable properties may be used provided

a They can be replaced by the standard parts (MS or AN) without alteration

b The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor drawings

3.2 2 Standard parts With the exception in 3 2 1, MS and AN standard parts shall be used where they suit the purpose They shall be identified on the drawings by their part numbers

3 3 MATERIALS

3 3 1 Fungus proof materials materials that are nutrients for fungi shall not be used where it is practical to avoid them Where used, and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary

3.3.2 Nonmagnetic materials Nonmagnetic materials shall be used for all parts except where magnetic materials are essential

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3.3.3 Metals Metals shall be of the corrosion-resistant type, or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service

3 3 4 Protective treatment When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification The use of any protective coating that will crack, chip or scale with age or extremes of climatic and environmental conditions shall be avoided

3 4 DESIGN AND CONSTRUCTION

3 4 1 Design The transmitter shall be designed with output ratios as specified in, and shall conform to MS24550 The ratios are to be achieved electrically Gears shall not be used The design shall be reasonably simple to permit overhaul or repair without requiring special tools and fixtures

3 4 2 Movement ratio The transmitter shall be designed to provide 180° of electrical output for each 90° shaft rotation for the 2 1 ratio and 90° shaft rotation for the 1 1 ratio transmitter The transmitter shall be designed to operate through 360° shaft rotation

3 4 3 Operation The transmitter shall be designed for either clockwise or counterclockwise operation For clockwise indicator pointer travel with clockwise transmitter shaft rotation as viewed from the shaft end, the external wiring shall be as shown on figure 1 For reverse operation leads A and C shall be interchanged

3 4 5 Construction The transmitter shall be so constructed that no parts will work loose in service and shall be built to withstand the strains, jars vibration and other conditions incident to shipping storage, installation and service without failure

3 5 Performance The transmitter shall be capable of meeting the test specified herein when subjected to the following conditions

a Temperature Temperature ranging form -65°C to 121°C ± 2°C, and exposure to +232°C ± 2°C for 10 minute cycles

b Scale error When operated over a 360° range scales errors not exceed ±2°

c Torque Torque required for operation not to exceed 0 5000 pounds-inch

d Insulation leakage When a potential of 100V dc is applied between any electrical pin contact or any metal part of the transmitter case

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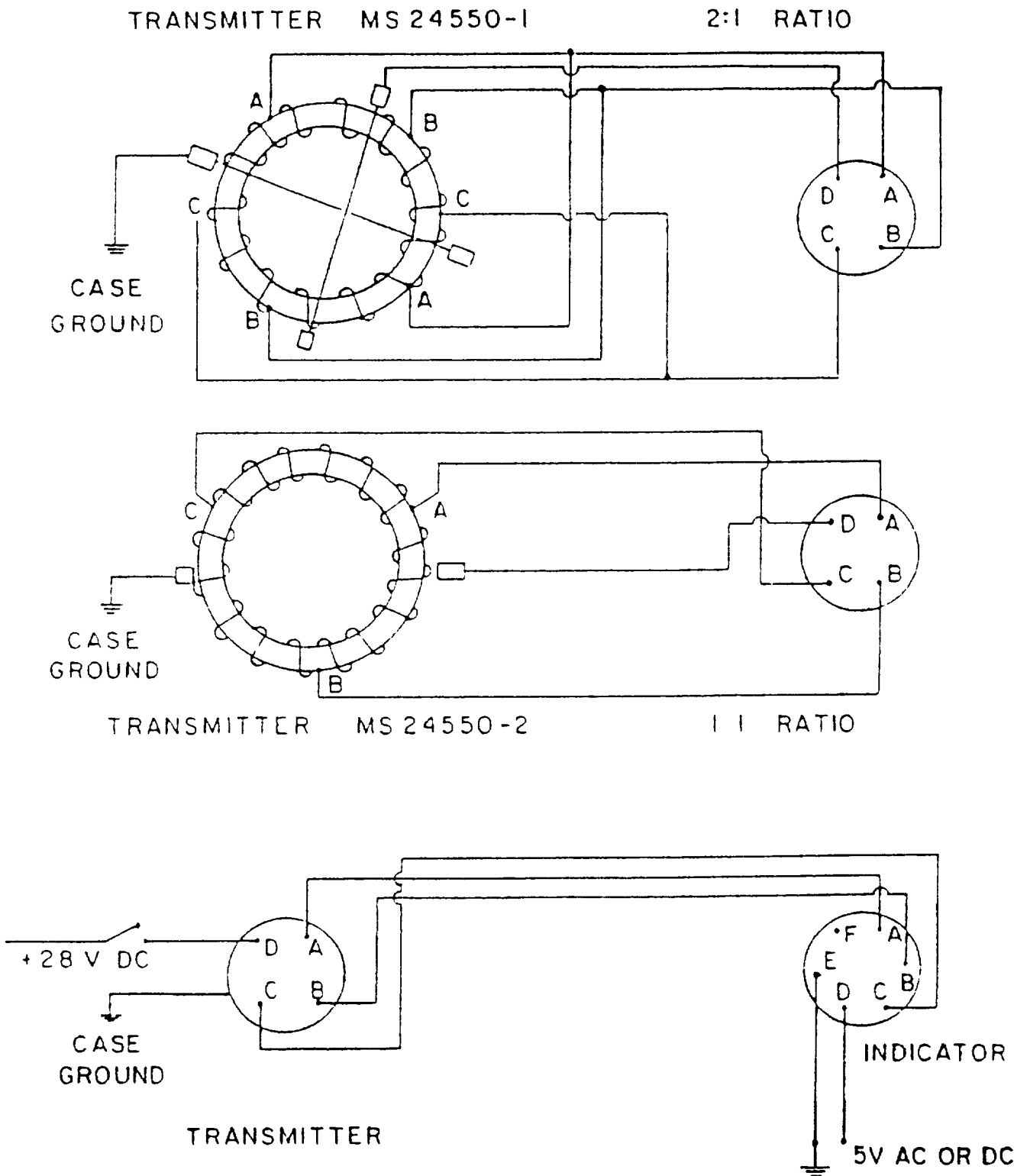


FIGURE 1 External wiring diagram.

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e Magnetic effect When revolved about a short bar magnet compass with a field intensity of 0 17 to 0 19 oersted

f Voltage variation A change in indication not to exceed 2° when voltage is varied from 22 500V dc to 30V dc

g Vibration Subjected to vibration at amplitudes, frequencies and times duration as specified in MIL-STD-810 Method 514 2 Category b 1, Curve B

h Salt fog Exposure to simulated salt sea atmosphere for a period of 50 hours

I Dust (fine sand) Exposure to sand and dust that may be encountered in desert areas

j Humidity Exposure to humidity cycling with up to 100 percent humidity

k Endurance When operated over a 360° range of period of 200 000 cycles

l Altitude Pressures ranging from 30 inches Hq down to 0 82 inch Hq (approximately 80,000 feet altitude)

3 6 Interchangeability All parts having the same manufacturer's part number shall be functionally dimensionally interchangeable The drawing number requirements of MIL-T-31000 shall govern changes in the manufacturer s part numbers

3 7 Sealed transmitter case The case shall provide a sealed enclosure for the entire internal mechanism may be removed form the case, replaced, and the case resealed This shall be accomplished without the use of special tools and fixtures unless they are approved by the contracting activity The sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the transmitter may be subjected Design drawings shall be submitted to the contracting activity for approval prior to fabrication of the transmitter case The transmitter case shall be made of nonferrous low-density metal uniform in texture with a smooth surface and covered with a durable dull black finish

3 7 1 Filling medium The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0 0006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure The filling medium shall be either 100 percent helium or a mixture of 88 percent to 92 percent nitrogen and the remainder helium The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere Where practical the 100 percent helium filling medium shall be utilized

3 8 Outline dimensions The outline dimensions of the transmitter shall conform to MS24550 The maximum dimensions shown are considered limiting dimensions and every effort shall be made to deep the size and weight as small as possible

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3.9 Electrical connector The transmitter shall be provided with four pin integral mounting receptacle as shown on MS24550

3 10 Wiring the internal wiring shall conform to figure 2. The transmitter shall be so wired that clockwise rotation of the shaft, as viewed form the shaft end, shall move the brushes from Tap A to Tap B to Tap C, et cetera The internal connections to the electrical receptacle shall be of flexible stranded wire

3 11 Resistance The direct current resistance s of the transmitter measured between the connector pins A to B, B to C and C to A shall be matched within 1 percent and have a resistance of 600 ± 150 ohms

3 11 1 Resistance values Deviation from the average of the resistance values between the three coil sections shall not exceed ±1 percent

3 11 2 Resistance windings The resistance windings shall be coated with a suitable cement, except at the line of contact to prevent displacement of the winding The contact surface of the winding shall be smooth and free of surface irregularities

3 12 Power The power for operation of the transmitter shall be 28V dc

3 13 Screw threads The threads of all machine screws, 0 060 inch diameter or larger, shall conform to MIL-S-7742

3 14 Safety wiring All external screws shall be secured by lockwiring in accordance with MS33540 or by an equivalent method of position retention

3 15 Lubrication Lubrication if used, shall be sufficient for approximately 1000 hours of use

3 16 Weight The weight of the transmitter completely assembled, less mounting nuts and mounting screws, shall not exceed 0.25 pound

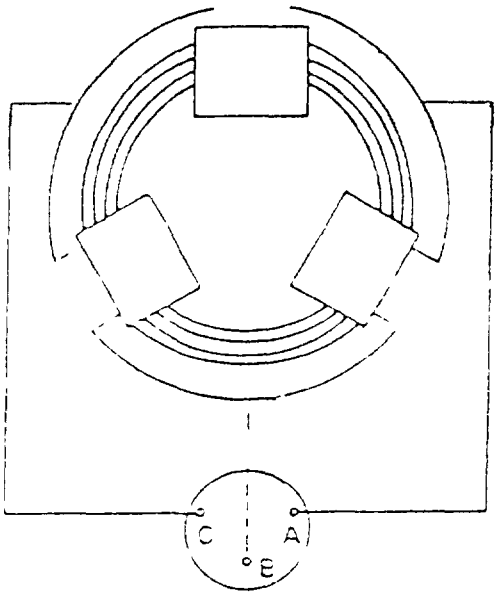
3 17 FINISHES AND PROTECTIVE COATINGS

3 17 1 Finish Protective coatings and finishes that will crack chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used

3 17 2 Aluminum alloy parts Aluminum alloy parts, where practicable, shall be covered with an anodic-film conforming to MIL-S-8625

3.17 3 Steel parts Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required Other protective coating, in lieu of MIL-

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RESISTANCE A-B, B-C AND A-C MATCHED TO ± 1% AND WITHIN 600 ± 150 OHMS

FIGURE 2 Internal wiring test indicator

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C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity Cadmium plating must be avoided when satisfactory alternative processes can be used.

3 18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3 19 INSTALLATION

3 19 1 Mounting screws The contractor shall finish sufficient mounting screws for installing the instrument They shall be No 6-32 NC-2A round head brass machine screws having durable dull black finish

3 19 2 Mounting nuts The transmitters shall be furnished with No 6-32 NC-2B thread self-locking insert nuts fitted to the mounting lugs

3 19 3 Envelope An envelope furnished by the contractor containing mounting screws shall be packed with each instrument, and shall be marked with the following information

"IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS"

3 20 Workmanship The transmitter shall be constructed and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454, requirement 9 Particular attention shall be given to neatness and thoroughness of soldering wiring, marking of parts and assemblies, plating painting, riveting, machine screw assembly welding, brazing, and freedom of parts from burrs and sharp edges

3 20 1 Dimensions Dimensions and tolerances not specified shall be as close as is consistent with best shop practices Where dimensions and tolerances affect the Interchangeability, operation or performance of the transmitter, they shall be held or limited accordingly

3 20 2 Screw assemblies Assembly screws and bolts shall be tight The word "tight" means the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads

3 20 3 Riveting Riveting operations shall be carefully performed to ensure that the rivets are tight and satisfactory headed

3 20 4 Gears Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots and other irregularities Where required, for accuracy adjustment, gear assemblies shall be free from backlash

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3 20 5 Cleaning The transmitter shall be thoroughly cleansed of loose, spattered or excess solder, metal chips or other foreign material after assembly Burrs and sharp edges as well as resin flash which might crumble shall be removed

3 21 Recycled and reclaimed materials. Recycled and reclaimed materials shall be used to the maximum extent possible without jeopardizing the end use of the item

4 QUALITY ASSURANCE PROVISIONS

4 1 Classification of inspection The inspection specified herein are classified as follows

a Acceptance inspection (see 4.2)

b First article inspection (see 4 5)

4 2 Acceptance inspection Acceptance inspections shall consist of

a Individual tests (see 4 2 1)

b Sampling plans and tests (see 4 2 2)

4 2 1 Individual tests Each transmitter shall be subjected to the following tests

a Examination of product (see 4 4 1)

b Torque (see 4 4.2)

c Operation (see 4 4.3)

d Scale error at room temperature (see 4 4 4)

e Voltage variation (see 4 4 5)

f Insulation (see 4 4 6)

4 2 2 SAMPLING PLANS AND TESTS

4 2 2 1 Sampling Plan A One transmitter selected at random from each 100 or less on the contract or order shall be subjected to the following tests

a Individual tests (see 4.2.1)

b Torque and scale error at low temperature (see 4.4.7)

c High temperature (see 4 4 8)

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d Vibration (see 4 4.9)

4 2.2 2 Sampling Plan B Unless otherwise specified, three transmitters selected at random from the first 15 items of the contract or order shall be subjected to the following tests.

a Sampling plan A tests (see 4 2.2 1)

b Transmitter life (see 4 4 10)

c High altitude-Low temperature (see 4 4 11)

d High temperature exposure (see 4 4 12)

e Fungus resistance (see 4 4 14 1)

f Salt fog (see 4 4 14 2)

g Dust (fine sand) (see 4 4 14 3)

h Humidity (see 4 4 14 4)

i Vibration (see 4 4 9)

j Magnetic effect (see 4 4 15)

k Mounting nuts (see 4 4 16)

l Mounting lugs (see 4 4 17)

4.2.2.2 1 Rejection and retest When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined

4 2 2 2 1.1 Individual tests may continue For operational reasons, individual tests may be continued pending the investigation of a sampling test failure But final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of the specification

4 2.3 Defects in items already accepted The investigation of a test failure could indicate that defects may exist in items already accepted If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them

4 3 Inspection conditions Unless otherwise specified all inspections shall be performed in accordance with the test condition specified in 4.3 1 of this specification.

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4 3 1 Standard atmospheric conditions Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29 92 inches Hq) and at room temperature (approximately +25°C) When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition

4 3 2 Tapping Unless otherwise specified in the tests, it shall be a three-coil, three wire position indicator whose scale errors are known in terms of the average production of the manufacturer The indicator scale shall be marked in 360 mechanical degrees (standard calibrating dial) The internal wiring of such an indicator is shown on FIGURE 1 Readings obtained shall be corrected for scale errors of the test indicator

4 4 TEST METHODS

4 4 1 Examination of product Each transmitter shall be examined to determine conformance with this specification with respect to materials, workmanship, and marking

4 4 2 Torque The torque required to turn the transmitter shaft shall be determined for clockwise and counterclockwise rotation The torque required shall not exceed 500 pounds-inch

4 4 3 Operation The transmitter shall be connected to a test indicator and a 28V dc source as shown on FIGURE 1 The indicator pointer shall move smoothly and uniformly throughout its range of travel as the transmitter shaft is rotated

4 4 4 Scale error at room temperature The transmitter shaft shall be rotated so that the indicator pointer is at the zero position For MS24550-1 transmitters the shaft shall be rotated in 30° increments to 180° and return, the MS24550-2 transmitters shall be tested at 60° increments The indication shall be noted at each 60° graduation from 0° to 360° and return from 360° to 0° The error shall not exceed ± 2°

4 4.5 Voltage variation The voltage applied to the transmitter shall be reduced to 22 500V with the indicator pointer at the 45° indication The voltage shall then be increased to 30V The change in indication shall not exceed 2°

4 4 6 Insulation An electromotive force of 100V dc shall be applied between the individual electrical pins and metal parts of the transmitter case A minimum resistance of 20 megohms shall exist This test shall be conducted during assembly and prior to grounding of wiring to case

4 4 7 Torque and scale error at low temperatures The transmitter and test indicator shall be connected as shown on FIGURE 1 The transmitter only shall be placed in a chamber, the

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temperature of which is controllable The temperature of the circulating air shall be reduced to -54° ± 2°C The transmitter shall be saturated at the low temperature for a minimum period of 4 hours The torque required to turn the transmitter shaft at this temperature shall not exceed 5 pound-inch The scale error test shall be conducted with the instrument at the specified temperature in the chamber The scale error at the low temperature shall not exceed ±4°

4 4.8 High temperature The transmitter shall be subjected to the high temperature test of +121° ± 2°C for a period of 4 hours The transmitter shall then be tested at the high temperature for scale and friction errors The scale error shall not exceed ±4°

4 4.9 Vibration The transmitter shall be operated to produce any indication between 30° and 60° and subjected to a vibration test in accordance with MIL-STD-810, Method 514.2, Category b 1, curve B The indication shall be determined without vibration and as the vibration frequency is varied Vibration of the transmitter shall not produce a pointer oscillation, or variation of amplitude exceeding 2° indication The change in indication after vibration test compared to the indication before vibration shall not exceed 2 indicated degrees

4 4 10 Transmitter life The transmitter shall be operated for 200.000 revolutions in which the brushes travel over the resistance winding 360° The initial 30 percent of the test shall be performed at 121°C and the final 70 percent at 20°C while the transmitter is operating at approximately 300 rpm After completion 200.000 cycles the scale error at room temperature shall be repeated The scale error shall not exceed ± 3° The requirements specified in 3.11 shall be repeated, with the deviation in resistance between sections not to exceed ±2 percent At the completion of these tests, the instrument shall be examined No physical damage shall be found as a result of this test

4 4 11 High altitude-Low temperature The transmitter shall be placed in a chamber maintained at a temperature of -65° ± 2°C and subjected to a pressure altitude condition of 80.000 feet (0 82 inches Hq) (absolute pressure) for a period of 4 hours Following this period, the transmitter shall be maintained at -65° ± 2° C at standard atmospheric pressure for 44 hours While at this temperature, the transmitter shall be operated throughout its range without exhibiting sticking or erratic characteristics The temperature shall then be raised to -55° ± 2°C for a period of 24 hours, and the transmitter shall meet the requirements for torque and scale error at low temperature

4 4 12 High temperature exposure The transmitter shall be subjected to a temperature of +121° ± 2°C for a period of 4 hours The temperature shall then be raised to 232° ± 2°C for 10 minutes This cycle shall be repeated 6 times for a total of 24 hours at the stabilized temperature of 121° and 1 hour at the stabilized temperature 232°C The transmitter shall be tested for scale error at +121°C Aster this first 4 hours of high temperature exposure and at the completion of the high temperature exposure. The last test shall be made while the transmitter is at +121°C. The scale error shall not exceed ± 4°

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4 4 13 Torque at high temperature The torque required to turn the transmitter shaft at high temperature shall not exceed 5 pounds-inch

4 4 14 Environmental The following tests shall be conducted in accordance with the specified procedures of MIL-STD-810C and as specified herein.

4 4 14 1 Fungus resistance Method 508 1, the unit selected shall not have previously been subjected to the salt fog test At the end of the test period the transmitter shall be carefully examined to ascertain that no fungus growth has occurred

4 4.1 4 2 Salt fog Method 509 1, for a period of 50 hours External connection shall be made to the equipment in such a manner as to stimulate installed conditions At the end of the 50-hour period, the equipment shall be subjected to and meet the individual tests specified herein

4 4 14 3 Dust (fine sand) Method 510 1 External connections to the transmitter shall be made to stimulate installed conditions At the conclusion of the test the equipment shall be examined and operated There shall be no evidence of sand or dust in the interior of the case that will affect subsequent operation of the transmitter

4 4 14 4 Humidity Method 507 1 Procedure I and as follows One sample transmitter mounted with its axis in a horizontal position shall be subjected to the humidity cycling procedure At the completion of the test the transmitter shall be examined for collection of moisture in the interior of the case for corrosion of metal parts, or for any other damage A second sample transmitter with case removed shall be subjected to the humidity cycling, and inspection for the same features shall be conducted with the exception of collection of moisture within the case The scale error at room temperature test shall be conducted prior to the cycling procedure The scale error at room temperature test shall again be conducted at the completion of the 10 cycles, and the change in indication shall not exceed the initial scale error at room temperature test by more than 2° There shall be no excessive corrosion of any of the components

4 4 15 Magnetic effect The transmitter shall be held in various positions with its nearest part 5 inches from and magnetically east or west of the center of a short bar magnet compass The horizontal magnetic field intensity shall be 0 17 to 0 19 oersted The compass shall be observed for change in deflection, which shall not exceed 3°

4 4 16 Mounting nuts Each attached mounting nut of the completely assembles transmitter case shall have an axial load of 25 pounds applied so as to tend to pull the nut out of the lug toward the rear of the case. There shall be no loosening of the nut A torque of 10 pound-inches shall then be applied and the rotation of the nut shall be limited by contact between a flat of the insert nut and the case.

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4 4 17 Mounting lugs The transmitter case, with the mechanism removed, shall be mounted in a suitable testing machine with the diameter of the case in a horizontal plane to ensure that the mounting lugs receive no added support A suitable pin shall be inserted through the hole in the mounting lug or the mounting nut and attached to a pull strap in the machine A load of 175 pounds shall be applied to each lug for a period of 1 minute along the mounting hole axis and toward the front of the case There shall be no fracture

4 5 FIRST ARTICLE INSPECTION

4 5 1 First article inspection samples The first article inspection samples shall consist of three transmitters Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization

4 5 2 First article required Prior to actual procurement the product which this specification covers shall pass the first article inspection specified herein If the product is alter modified in any way, the modified form shall be subjected to and shall pass the same first article inspection

4 5 3 First article tests The first article tests shall consist of all the test specified herein (4 4)

5 PACKAGING

5 1 Preservation Preservation shall be level A, C or industrial. IAW MIL-STD-2073 1A. as specified (see 6 2)

5 1.1 Level A.

5 1 1 1 Cleaning Transmitter shall be cleaned in accordance with process C-1 or MIL-P-116.

5 1 1 2 Drying Transmitter shall be dried in accordance with process D-4 of MIL-P-116

5 1 1 3 Preservation application Preservatives shall not be used

5 1 1 4 Unit packaging Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method 1A of MIL-P-116 Each transmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant with sufficient cushioning material to ensure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subject to the rough handling drop tests of MIL-P-116

5 1 2 Level C Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

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5.1 3 Industrial The industrial preservation of transmitter shall be in accordance with ATM D3951

5.2 Packing Packing shall be level A, B, C, or industrial as specified (see 6 2)

5 2 1 Level A Transmitter packaged as specified in 5 1 1 shall be packed in shipping containers conforming to PPP-B-601, Style A or B, Class overseas, unless otherwise specified by the contracting activity Insofar as practical exterior shipping container shall be of uniform shape size minimum tare and cube consistent with the protection required

5 2 2 Level B Transmitter packaged as specified in 5 1 1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant unless otherwise specified by the contracting activity Other requirements as specified in 5 2 1 apply

5 2 3 Level C Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use This level shall conform to applicable carrier rules and regulations

5 2 4 Industrial The packaged transmitter shall be paced in accordance with ASTM D3951

5 3 Marking In addition to any other markings required by the contract or order (see 6 2) interior and exterior containers shall be marked in accordance with MIL-STD-129

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory)

6 1 Intended use The transmitter covered by this specification is intended for use in aircraft to transmit electrically the relative positions of control surfaces in degrees to a remote indicator

6 2 Acquisition requirements Acquisition documents must specify the following

a Title, number, and date of this specification

b MS Part No of transmitter desired (see 1 2)

c Reinspection date markings

d Level of packaging and packing desire

e Whether sampling plan B tests are to be conducted

f Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents reference (see 2 1)

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Requisition, contract, and orders should specify the part numbers of the transmitter which are to be procured, and whether overseapacking is desire.

6.3 DEFINITIONS

6.3 1 Tapping The phrase TAPPED LIGHTLY is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments

6 3 2 Amplitude Whenever the work AMPLITUDE is specified, it means the extent of motion as measured from one extreme to the opposite extreme

6 4 First article A first article is required for inspection and approval (see 3 1, 4 1, and 4 5)

6 5 Subject term (key word) listing

DC
Position
Transmitter
VDC
IOA

6.6 Changes from previous issue Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Custodian
AIR FORCE - 99

Preparing Activity
AIR FORCE - 71

Project Number
6610-F388

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-T-26609C	15 OCTOBER 1994

3 DOCUMENT TITLE TRANSMITTER, POSITION

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code (1) Commercial (2) AUTOVON (If applicable)	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME HARLENA EDWARDS	b TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON 336-5960
c ADDRESS (Include Zip Code) OC-ALC/TICLA 3001 Staff Drive Tinker AFB, OK 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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